Spurious Predictability from an Embedding Index Error: A Case Study in Financial Analog Forecasting

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Abstract

We document and trace to its root cause a case of spurious predictability in delay-embedding-based financial forecasting. Testing four novelty/anomaly detectors across seven asset classes, we initially found forecast accuracy substantially higher on atypical days – a result that survived walk-forward cross-validation, permutation testing, and a genuine out-of-sample holdout. It was an artifact of a one-step index error: the embedding's most recent component was numerically identical to the first step of the future return being predicted. We confirm this diagnosis with an algebraic proof, a pipeline-free correlation test, a null control on synthetic white noise, and a power test showing the corrected pipeline still detects genuine designed-in signal; all four detectors were affected identically, pointing to the shared embedding rather than any one method. We propose a null-noise-plus-surrogate-data protocol that would have caught this error in hours rather than months, and recommend it as standard practice before reporting a positive result in this class of study.

Keywords: spurious predictability, data leakage, delay embedding, analog forecasting, surrogate data, anomaly detection

1. Introduction

Short-horizon financial return forecasting via historical-analog methods – delay-embed recent returns into a low-dimensional state space, locate the k nearest historical analogs, and average their realized continuations – is a long-standing technique with roots in the Analog Ensemble literature from atmospheric and space-weather forecasting [4] and in the broader nonlinear-dynamics tradition of state-space reconstruction [14, 9]. A natural extension,

explored across a large and active recent literature, is to condition forecast confidence or forecast usage on an auxiliary novelty- or anomaly-detection score computed on the same embedded state: the intuition is that "unusual" market states might be more (or less) predictable than "typical" ones.

This paper began as an attempt to formalize and validate exactly that intuition. Using a diffusion-model reconstruction score (and, for robustness, three classical novelty detectors: Isolation Forest, Local Outlier Factor, and Mahalanobis distance in PCA space) as the atypicality signal, we found – across seven asset classes (equity index, two energy benchmarks, two precious metals, two cryptocurrencies) – that forecast accuracy conditional on high atypicality was substantially and significantly higher than conditional on low atypicality. The result replicated under nested walk-forward evaluation, combinatorial purged cross-validation, formal permutation testing [7], a genuine single-shot 20% holdout that was never touched during model development, and even a proposed dynamical-systems explanation involving local Lyapunov exponents.

It was wrong. Section 4 describes how we found out, and why none of the validation steps listed above caught it. We believe the diagnostic protocol we used to find and confirm the error – and which we recommend other researchers adopt – is the paper's main contribution.

1.1. Contributions

1. We document a previously-unreported class of index error in delay-embedding-based forecasting pipelines, under which the embedding's most recent feature is numerically identical to part of the forecast target, and show it survives methods normally considered strong safeguards against overfitting (walk-forward CV, permutation testing, genuine out-of-sample holdout).
2. We verify the diagnosis at six increasing levels of pipeline complexity, from a closed-form algebraic proof to a null-hypothesis control on synthetic data with a provably known absence of structure – noting explicitly that this is six checks at different levels of abstraction, not six independent sources of evidence (checks 1 and 3 are two representations of the same underlying fact).
3. We show the error affected all four structurally different detectors tested identically, establishing that the apparent "finding" was a property of the shared embedding convention, not of any one detection method.

4. We propose a lightweight, two-part diagnostic protocol (a pure-noise null control, and AAFT surrogate-data testing) that would have caught this error immediately, and that we recommend as a standard pre-submission check for this entire class of method.

2. Related Work

Delay embedding and analog forecasting. The analog-forecasting technique underlying this investigation – locate the k nearest historical states to the current embedded state and pool their verifying continuations – is structurally identical to the Analog Ensemble (AnEn) framework developed in the atmospheric-forecasting literature. Hamill and Whitaker [8] introduced the underlying analog-correction technique for precipitation reforecasts; Delle Monache et al. [4] extended it with a Kalman-filter/analog post-processing scheme, and Delle Monache et al. [5] formalized the probabilistic AnEn method as a way of generating an ensemble forecast directly from a single deterministic trajectory. Alessandrini et al. [1] demonstrate AnEn's practical value for short-term wind-power forecasting, illustrating the framework's reach beyond its original precipitation application. The state-space reconstruction underlying the embedding itself follows the standard delay-coordinate approach of Takens [14] and the broader nonlinear time-series methodology surveyed in Kantz and Schreiber [9]. To our knowledge, the present investigation (and the broader project it is drawn from) is the first to apply the AnEn framework, under that name, to financial return series rather than to a meteorological or physical system.

Spurious predictability and leakage in financial machine learning. The closest precedents to this paper's diagnostic contribution are Nikolopoulos [11], "Spurious Predictability in Financial Machine Learning," which formalizes a falsification framework for diagnosing exactly this class of problem in financial ML pipelines; the foundational survey on leakage formulation, detection, and avoidance in data mining more broadly [10]; and a 2025/2026 preprint demonstrating that decomposition-ensemble cryptocurrency forecasting models' reported directional accuracy is attributable almost entirely to look-ahead bias inherent in two-sided signal-decomposition methods (CEEMDAN/VMD), which proposes a falsification test structurally similar in spirit to the null-noise control used here (Research Square preprint, DOI: 10.21203/rs.3.rs-11529129/v1; author to be confirmed before final submission). On the statistical side, the multiplicity concerns raised by testing

many model/detector variants on the same panel are formalized by White [17]'s Reality Check for data snooping and by the "garden of forking paths" framework of Gelman and Loken [6] and its direct application to financial-anomaly research by Coqueret [3].

Surrogate-data testing and the chaos-in-markets debate. Our diagnostic protocol's second component draws directly on the surrogate-data methodology introduced by Theiler et al. [16] for testing nonlinearity against a matched-linear null, and refined by Schreiber and Schmitz [12, 13]'s Iterative Amplitude-Adjusted Fourier Transform (IAAFT), a closer-to-exact refinement of the AAFT surrogates used in this paper. This methodology underlies the decades-long, still-unsettled literature testing whether financial return series exhibit genuine deterministic chaos rather than linear stochastic structure that merely resembles it (surveyed critically in Barnett and Serletis [2]); studies applying surrogate-data and related nonlinearity tests to individual equity indices have reached conflicting conclusions depending on denoising and test choice, underscoring that this is an active methodological debate rather than a settled question. This literature's central lesson – that apparent nonlinear structure in a detector statistic must be checked against a matched linear surrogate null before being attributed to genuine dynamics – is the lesson this paper's own investigation initially failed to apply to its *own* core predictability claim, not only to its exploratory dynamical-systems explanation.

3. Data and Baseline Method

3.1. Data

Daily close prices for seven assets, downloaded from Yahoo Finance: Brent crude (`BZ=F`), the S&P 500 index (`^GSPC`), gold (`GC=F`), silver (`SI=F`), WTI crude (`CL=F`), Bitcoin (`BTC-USD`), and Ethereum (`ETH-USD`), using the maximum available history per asset (8–20 years depending on listing date).

3.2. Delay embedding and analog selection

Log returns $r_t = \log(P_t/P_{t-1})$ are delay-embedded into $\mathbb{R}^M$ with $M = 3$, $\tau = 1$: $x_j = (r_t, r_{t-\tau}, r_{t-2\tau})$. For a forecast origin, the $K = 7$ nearest historical analogs (by a learned feature distance; see Section 3.3) are selected causally, excluding a Theiler window of 10 days around the origin to avoid trivial temporal auto-matches [15]. The forecast is the sign of the mean $H = 3$-day forward return realized by the selected analogs.

3.3. Atypicality detectors

Four detectors score each origin's "atypicality": (i) the reconstruction loss (ELBO) of a small denoising diffusion model (a 2-layer MLP, hidden width 32) trained causally on the embedded history; (ii) Isolation Forest; (iii) Local Outlier Factor; (iv) Mahalanobis distance in a PCA-reduced space with Ledoit-Wolf shrinkage. All four are fit causally, retrained periodically on history up to the current origin only.

3.4. Evaluation

Nested nonoverlapping walk-forward folds (step size H); a 2000-resample permutation test for the accuracy-gap statistic (median-split by atypicality score); Fisher's method to combine p-values across the seven-asset panel; and, separately, a single untouched 20% holdout per asset, evaluated exactly once.

4. The Error: Discovery, Diagnosis, and Fix

4.1. The original result

Using the method of Section 3, we found a substantial, statistically significant gap in forecast accuracy between high- and low-atypicality origins: for example, the diffusion-ELBO detector showed a +8.3 percentage-point accuracy gap on Brent crude ($p < 0.0001$), and three classical detectors (Isolation Forest, LOF, Mahalanobis-PCA) showed comparable or larger gaps on the same and other assets (Table 1 reports the full pre-fix comparison across four detectors and seven assets). A genuine single-shot holdout test, touched for the first time after all model development was frozen, reproduced the effect at the panel level (Fisher-combined $p = 3.8 \times 10^{-5}$ across seven assets). We additionally proposed and validated a dynamical-systems explanation (local Lyapunov exponents), which itself replicated on a second, independent holdout and was robust to parameter choices and to splitting the sample by calendar era.

4.2. The surrogate-data control that exposed the problem

Motivated by a standard robustness check for the dynamical-systems explanation specifically (not yet the core claim), we generated AAFT surrogate return series – matching the real data's amplitude distribution and linear power spectrum exactly, with phases randomized to destroy any nonlinear

temporal structure – and re-ran the local-Lyapunov analysis. The purely linear surrogates reproduced (and in fact exceeded) the real data's effect size, for both a cheap k-NN atypicality proxy and the production diffusion-ELBO detector. This result, on its own, is explicable without positing any coding error (surrogate-data literature is full of "apparent nonlinear structure is actually linear" findings) – but it prompted us to apply the same control to the paper's actual central claim: the atypicality-conditioned *accuracy* gap, not merely the auxiliary Lyapunov statistic.

4.3. The decisive control: pure white noise

We generated literal i.i.d. Gaussian white noise – a process with, by construction, zero genuine forecastable structure of any kind – and ran the identical pipeline on it. The pipeline reported a "significant" atypicality-conditioned accuracy gap of 5–7 percentage points, with permutation p-values as low as the resolution floor of a 2000-resample test (0.0000), on every one of three initial noise draws (a result extended to ten trials across two assets in Section 5, with the same conclusion). A process with zero genuine predictive structure, by construction, cannot produce a genuine 5–7 point accuracy edge; this result can only be explained by a defect in the measurement procedure itself.

4.4. Root cause

In plain terms: the pipeline's "most recent known return" – the feature used to judge how unusual today looks and which historical days resemble it – was not actually known yet at the moment the forecast was made. It was one day into the future the model was supposedly predicting.

More precisely: with origin index j, embedding length M, and delay τ, the project's convention located "today's price" at array index $p = j + (M-1)\tau$ and defined the forecast target as the H-step return beginning at p. But the embedding's most recent component, $x_j^{(0)}$, is constructed from $\texttt{log_ret}[j + (M-1)\tau]$ – exactly $\texttt{log_ret}[p]$: the **first of the H returns summed to produce the forecast target itself**. The correct convention is $p = j + M\tau$, which places the embedding entirely in the past relative to the forecast window, with zero gap and zero overlap.

4.5. Why none of the standard safeguards caught it

The error sits in every fold's feature-construction code identically, in both the "training" and the "held-out" portions of the data – it is a fixed, deterministic property of the index arithmetic, not something learned from or overfit

to any particular sample. This is precisely why each of the three safeguards this literature relies on most heavily missed it:

- **Walk-forward cross-validation** protects against overfitting to one historical window. It offers no protection against a structural bias present identically in every window.
- **A genuine single-shot holdout** protects against re-using information learned during model selection. It offers no protection against a bias baked into the feature-construction code itself, which is applied, by construction, the same way to the holdout period as to everything else.
- **Permutation testing** protects against mistaking sampling noise for a real effect. It offers no protection when the "effect" is a deterministic artifact that reappears, with the same mechanism intact, in every permutation.

Each of these three tools is, individually, considered strong evidence in the forecasting literature – and none of them is designed to catch a fixed arithmetic bias in feature construction, because none of them varies the feature-construction code itself. A fourth tool does: comparing the pipeline's output on real data against its output on data engineered to contain no real structure at all (Section 4.3; formalized as a general protocol in Section 7.2) directly tests the feature-construction code, not the sample.

5. Verification

Because of the scale of what the diagnosis implies – the central claim of a multi-month investigation turning out to be an artifact – we did not accept it on the strength of any single check. The six checks below span increasing levels of pipeline complexity, from a closed-form algebraic identity to the full production pipeline; checks 1 and 3 are two representations of the same underlying fact rather than independent evidence, and are listed separately only because they operate at different levels of abstraction (full detail and numbers: README Findings 51–52 of the accompanying code repository).

1. **Exact algebraic identity**: hand-derived and numerically confirmed that $x_j^{(0)}$ equals `log_ret`$[p]$ exactly, not approximately, on a tiny synthetic price array.

2. **Generalization across the real series**: confirmed over 33 (resp. 47, in a fully independent re-implementation) origins spread across real Brent history: the overlap holds universally under the old formula and never under the corrected one.
3. **A pipeline-free, model-free direct test**: $\text{sign}(x_j^{(0)})$ matches sign(actual future return) in 68.3% of folds under the old convention versus 50.1% (chance) under the fix – 68.3% rather than ≈100% is itself consistent with the diagnosis, since the leaked component is only one of the three summands making up the target.
4. **A null control repeated ten times** across two assets and two random seeds: under the fixed pipeline, mean permutation p-value across trials was 0.496 (the theoretical expectation under a true null is exactly 0.5), with a false-positive rate matching the nominal 5% level.
5. **A fully independent, from-scratch script with zero imports from the project's own codebase**, using a fresh data download and hand-written embedding/index arithmetic, reproduced the identical result, ruling out a shared misreading of the code baked into the investigation's own tooling.
6. **A power test with a known, designed-in signal**: a synthetic series with an explicit, by-construction magnitude-conditional mean-reversion rule (after any return exceeding one standard deviation, the next three returns are deliberately biased in the opposite direction) was correctly detected by the fixed pipeline (accuracy 52.2%→58.2%, gap $p = 0.011$), confirming the fix did not simply desensitize the method.

We additionally ran a systematic six-point leakage audit beyond the specific bug (data-feature temporal ordering; embedding-normalization causality; train/test shuffling; target-feature alignment; residual embedding overlap; detector self-inclusion). One further, independent, and much smaller issue was found – the embedding's scale normalization is computed over the full series rather than causally – disclosed, and shown empirically not to materially affect the result (Brent: +2.4pp → −0.1pp under a corrected causal-scale variant, both non-significant; S&P 500: −1.2pp → −1.2pp, materially unchanged).

6. Results After Correction

No accuracy gap reaches significance; several point estimates are negative. Extending the comparison to all four detectors across the full seven-asset

Table 1: Diffusion-ELBO accuracy gap, corrected pipeline, by asset.

Asset	Accuracy gap	p-value
Brent	+2.4pp	0.176
S&P 500	−1.2pp	0.692
Gold	−5.4pp	0.985
Bitcoin	−1.7pp	0.720
Ethereum	+1.2pp	0.352

panel (28 tests; see README Finding 53 for the complete table) finds exactly 1 of 28 cells below $p = 0.05$ (Ethereum/Mahalanobis-PCA, $p = 0.025$) – indistinguishable from the ≈1.4 false positives expected by chance alone at $\alpha = 0.05$ across 28 independent tests.

The genuine single-shot holdout test, re-run with the corrected pipeline, tells the same story at the panel level: every asset is individually non-significant, and Fisher's combined p-value across the seven-asset panel is **0.288**, against the original (uncorrected) pipeline's $p = 3.8 \times 10^{-5}$ on the identical protocol – the strongest-looking evidence in the original investigation collapses to a result indistinguishable from chance.

7. Discussion

7.1. Why this is worth publishing

A null result reached through careful, multiply-verified diagnosis of a subtle and previously undocumented methodological error is, we argue, more valuable to the forecasting community than the original (incorrect) positive finding would have been, for three reasons. First, the error is structural to a widely-used class of method (delay-embedding plus auxiliary novelty/anomaly scoring), not specific to our implementation choices, and is therefore likely to recur in other studies using the same convention. Second, we show concretely that the standard toolkit of safeguards against overfitting in this literature – walk-forward validation, permutation testing, even genuine out-of-sample holdout testing – did not detect it, which is itself an actionable finding about the limits of those safeguards for this specific failure mode. Third, we provide a lightweight, generally-applicable diagnostic (a null-noise control plus surrogate-data testing) that would catch this class

of error cheaply, before a researcher invests months in building an elaborate positive-finding narrative on top of it, as we did.

7.2. A recommended diagnostic protocol

We recommend that any study using delay-embedding plus an auxiliary detection score for financial (or other) time-series forecasting run, prior to reporting a positive result, at minimum: (i) a null-data control – apply the identical pipeline to synthetic data with provably zero real structure (i.i.d. noise matching the real series' scale is sufficient) and confirm the reported effect does not appear; (ii) a surrogate-data control – apply the identical pipeline to AAFT (or comparable) surrogates of the real series, to separate genuine structure from an artifact reproducible by matched linear noise. Both checks are inexpensive relative to the modeling effort typically invested in this class of study, and both would have caught the error documented in this paper within hours rather than months.

7.3. Limitations

This paper documents one specific error, in one specific project's specific implementation of a standard method family. We do not claim the error is present in any other published study; we claim only that the general class of convention (using the embedding's most-recent-return component without an explicit, verified alignment check against the forecast target's own starting index) carries this risk, and that the diagnostic protocol we propose is a cheap, general-purpose safeguard regardless of whether any specific paper happens to contain this exact error.

8. Conclusion

We set out to validate a claim that market states flagged as locally atypical by an unsupervised novelty detector are more forecastable than typical states. The claim initially passed every standard validation step used in this literature. It did not survive a null-data control, because the apparent effect was in fact a one-step index error that gave the forecasting model partial access to its own target. We confirm this diagnosis at six increasing levels of pipeline complexity and show the corrected pipeline retains full power to detect genuine designed-in signal, ruling out that the correction merely broke the method. We offer the diagnostic protocol that exposed the error – a null-noise control and a surrogate-data test, both standard tools in adjacent

literatures but, to our knowledge, not yet standard practice in this specific subfield – as this paper's primary, generalizable contribution.

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